

Uncovering LLM’s Capabilities in Model-based Question Answering for UML Class Diagrams

Online Supplementary Material

Manuel Mischak¹, Charlotte Verbruggen¹^[0000–0003–0418–2633], Philip Langer²,
and Dominik Bork¹^[0000–0001–8259–2297]

¹ Business Informatics Group, TU Wien, Austria
`{manuel.mischak,charlotte.verbruggen,dominik.bork}@tuwien.ac.at`
² EclipseSource Services GmbH, Vienna, Austria
`planger@eclipsesource.com`

Abstract. In this document, the online supplementary material for the EMMSAD 2026 paper *Uncovering LLM’s Capabilities in Model-based Question Answering for UML Class Diagrams* is provided.

Keywords: Conceptual modeling · Model comprehension · LLM · Accessibility.

1 Systematic Literature Review

This document presents the literature review that was conducted to address the following research question:

RQ1: What is the state of the art in model comprehension and conversational agent techniques in conceptual modeling, and what are the agents’ strengths and limitations?

1.1 Search Queries

The search queries are designed to capture the literature at the intersection of model comprehension/summarization and accessibility, particularly focusing on auditory conversion techniques. To ensure comprehensive coverage of the literature, three interrelated queries were formulated. Each query targets literature at the intersection of model-related research in one of the following domains:

- **Q₁: Model Accessibility**
- **Q₂: Model Comprehension and Simplification**
- **Q₃: Conversational Chat Bots and Interfaces**

The main reason for this approach is to cover the full spectrum of research where model representations are either simplified or made more accessible, either through direct summarization/comprehension techniques or via conversational user interfaces.

The search queries are using the following logical structure:

Queries $Q_i = MC \wedge (D_i)$ *where*

MC stands for the Modeling Keywords, and D_j includes the keyword sets for the specific domain used for each query with D_1 relating to Model Accessibility, D_2 relating to Model Comprehension and Simplification, and D_3 relating to Conversational Chat Bots and Interfaces.

Since the Modeling Keywords form the foundation for all three searches, this section applies uniformly to each one. The keywords used for the MC set are as follows:

$MC \in \{ "UML" \vee "BPMN" \vee "Entity\ Relationship" \vee "Entity-Relationship" \vee "Conceptual\ model*" \vee "Semantic\ model*" \vee "Data\ model*" \vee "Software\ model*" \vee "Model\ Driven" \vee "Model-Driven" \}$

and

$D_1 \in \{ "Accessibility" \vee "Auditory\ conversation" \vee "Text-to-speech" \vee "Conversational\ agent*" \vee "Voice\ interface*" \vee "Visual\ impairment*" \vee "Accessible\ user\ interface*" \}$

and

$D_2 \in \{ "Model\ summar*" \vee "Model\ simplif*" \vee "Model\ comprehen*" \vee "Model\ understand*" \vee "Diagram\ summar*" \vee "Diagram\ simplif*" \vee "Diagram\ comprehen*" \vee "Diagram\ understand*" \}$

and

$D_3 \in \{ "Conversational\ agent*" \vee "Voice\ inter*" \vee "Accessible\ user\ inter*" \vee "Auditory\ conversation" \vee "Auditory\ inter*" \vee "Voice-based" \vee "Voice\ based" \vee "Text-to-speech" \vee "Chatbot" \vee "Chat-bot" \}$

By combining any keyword from MC with any keyword from D_3 using the logical AND operator, the search query targets literature that covers conversational aspects in modeling.

The search queries were conducted using prominent scientific databases, including Scopus, IEEE Xplore, and ACM Digital Library with a focus on Scopus. The queries were customized to match each database's syntax, ensuring it retrieved results from all categories like Title, Abstract, and Keywords of the literature. Additionally, the search was not limited to any specific publication years.

1.2 Exclusion Criteria

- **EC1:** Publications written before 2010
- **EC2:** Publications with less than 4 pages
- **EC3:** Publications not written in English
- **EC4:** Publications that are not related to the primary subject areas addressed in this paper
- **EC5:** Publications not accessible as full text or non-scientific papers

1.3 Screening Process

To ensure a thorough and methodical selection of literature, a multi-stage screening process was employed. In the initial first phase, the titles of all retrieved publications were evaluated, and those deemed clearly irrelevant to the research objectives were discarded. Subsequently, the abstracts of the remaining works were examined to verify their thematic alignment and potential contribution to the research questions. This step served to eliminate studies that lacked sufficient relevance or depth. Publications that passed the abstract-level screening proceeded to a full-text evaluation, where their content was critically assessed to confirm methodological soundness and thematic suitability. Finally, a quality assessment was carried out for each candidate publication, considering factors such as research design, sample size, and the robustness of findings. Following this comprehensive screening, the selected publications were organized into relevant thematic categories, forming a solid foundation for further analysis and integration into the research framework.

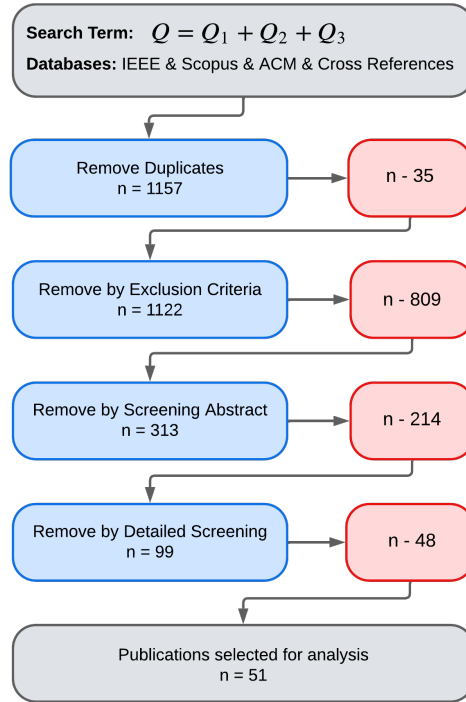


Fig. 1. Publications search process and filtering steps

As shown in Figure 1, the literature selection process can be visualized as a funnel, starting with a broad set of publications and progressively narrowing down to a core set of highly relevant works, that will be used in the next section.

1.4 Results of the Search Process

Initial Search and Compilation. The literature search began with broad queries across three key databases:

- **Scopus:** 961 publications
- **ACM Digital Library:** 82 publications
- **IEEE Xplore:** 95 publications
- **Cross References:** 19 publications

The total number of studies initially identified was 1,157. After merging the four sources, duplicates were identified and removed. Since each source individually contained no duplicates, the total number of removed publications following the merge was only 35.

Applying Exclusion Criteria. All duplicate cleared publications were subjected to the predefined exclusion criteria (Section 2.1.2). Reasons for exclusion included insufficient thematic relevance, language restrictions, and publication date constraints. This filtering step narrowed the set from 1,122 to 313 publications, by removing 809 studies.

Abstract-Level Screening. The abstracts of the remaining 313 publications were carefully reviewed to ensure alignment with the research objectives. At this stage, 214 publications were removed due to lack of relevance or insufficient depth, leaving a refined set of 99 publications for more detailed examination.

Thematic Categories. Before conducting the full-text review, each of the 99 shortlisted publications was mapped to thematic categories that reflect the key interests of this paper. These categories guided the assessment and helped ensure that the final selection would directly support the research objectives:

- **Accessibility:** Studies on inclusive design and assistive technologies aimed at making complex models usable by a diverse user base, including individuals with visual impairments.
- **Summarization and Comprehension:** Research focusing on means of condensing and abstracting, and comprehending complex models, thereby easing the interpretation and navigation of their contents.
- **Conversational:** Works exploring conversational agents or voice interfaces as intuitive, user-friendly channels for interacting with complex model information.

These categories align closely with the core research questions and objectives, ensuring that the final literature base would be directly relevant to the development and evaluation of the proposed prototype.

Full-Text Screening and Final Selection. Full-Text screening of the 99 shortlisted publications evaluated their methodological quality, clarity of contribution, and direct applicability to this paper. This comprehensive screening led to the exclusion of 12 publications, culminating in a final selection of 51 publications. These 51 studies were then classified according to their thematic scope.

This final set of 51 publications forms a thematically diverse yet focused foundation for subsequent analysis. These works inform the conceptualization phase, guide the prototype design, and support the evaluation efforts outlined in the following sections.

1.5 Summary and Results

In this section, we examine the publication output and categorization of the selected papers. This analysis not only highlights the growth in research over the years but also provides insight into how different studies contribute to various thematic areas.

Publications by Publication Year An examination of the retrieved documents categorized by publication year (see Figure 2) reveals a pronounced upward trend in the volume of relevant research over time, with a particularly steep increase in recent years. This surge aligns with the rise of new technological possibilities, notably the advent of advanced AI-driven solutions such as chatbots, which have substantially enhanced accessibility in model engineering. As these tools become more sophisticated and user-friendly, they not only lower the barriers to entry for researchers and practitioners but also expand the scope of potential applications. Consequently, the academic and industrial communities have responded with increased publication activity, reflecting both the growing interest in leveraging these innovations and the rapid evolution of the underlying technologies.

Search Hits per Query and Database In addition to the qualitative analysis presented above, a quantitative overview of the search process was conducted. Table 1 summarizes the number of hits retrieved per search query and per database. These initial hit counts provide insight into the extent of the literature captured by each query before applying further screening and exclusion criteria. Note that only for the first query all three databases were used and for Q₂ and Q₃ only Scopus was used as a source for publications.

Table 1. Hits per Query and Database

Query	Scopus	ACM	Digital Library	IEEE
Model Accessibility	379	82		19
Model Summarization and Comprehension	344	-		-
Conversational Interfaces	238	-		-

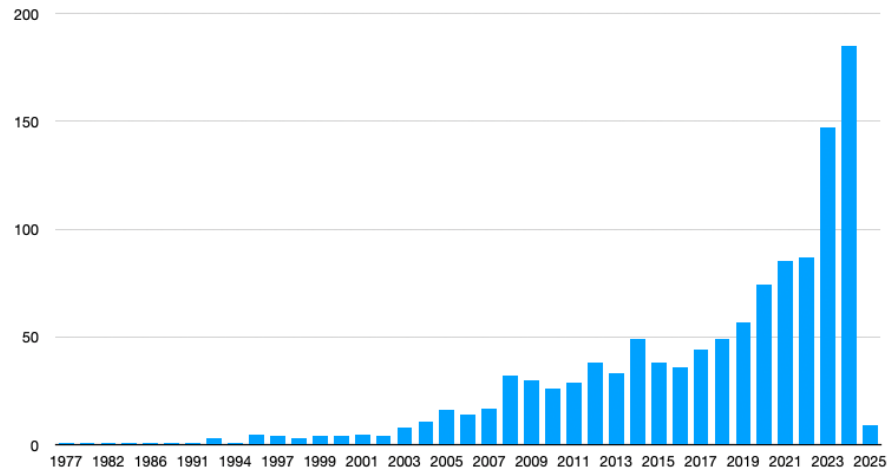


Fig. 2. Publications by publication year

Publications by Category The selected publications have been organized into thematic categories based on their main focus areas. The categories include Accessibility, Summarization and Comprehension, and Conversational interfaces. While some studies concentrate on a single theme, several publications address overlapping areas, reflecting the interdisciplinary nature of the research (see Figure 3). This categorization not only simplifies the overview of the literature but also lays the groundwork for deeper analysis in subsequent sections.

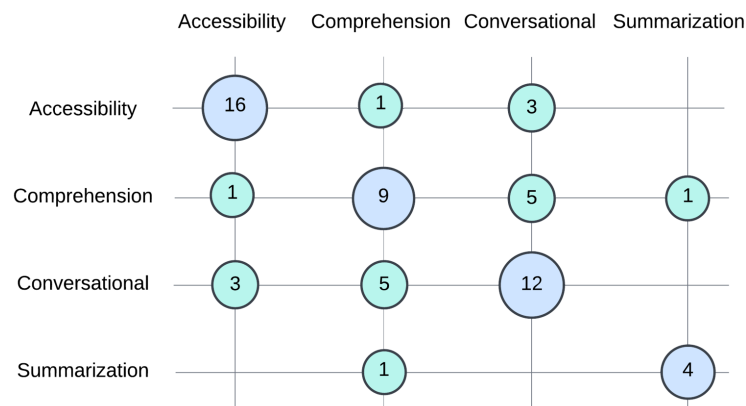


Fig. 3. Publications by category

Table 2 lists the most relevant references for each category. Each reference has been assigned to the category where its contribution is most significant, even if its content spans multiple areas. For this purpose, the topic of Summarization and Comprehension is split in two columns. This approach helps readers quickly identify key studies within each thematic area and understand how they specifically address different aspects of the broader research topic.

Table 2. Key references organized by topic.

Model Accessibility	Summarization	Model Comprehension	Conversational Interfaces
[26, 22, 35, 25, 19, 52, 39, 18, 7, 45, 53, 50, 44, 36, 1, 30, 46, 38, 27]	[4, 43, 10, 23, 31]	[16, 49, 40, 11, 14, 5, 8, 3, 48, 51]	[17, 2, 12, 33, 34, 28, 32, 20, 47, 29, 41, 24, 15, 9, 6, 42, 13]

2 Promptfoo Evaluation

To provide a concrete insight into how models and comprehension questions were presented in the test setup, the following illustrates one of the model-question pairs. Figure 4 presents the JSON representation of model q-E1 from the original empirical study by Sharif and Maletic [37], which was recreated in BIGUML without alteration.

The corresponding comprehension question is also sourced directly from [37]:

Which class is involved in a composition relationship with QAction and is part of the QEvent hierarchy?

It assesses whether the LLM can identify the correct class and interpret its semantic role in the inheritance and composition structure. The question was translated into the rubric configuration presented in Listing 1.1. This configuration applies the criteria for correctness and justification and follows the internal structure of promptfoo assertions.

Listing 1.1. Question 1 of q-E1

```

1 - vars:
2   uml: file://workspace/qt-E/qt-E-plantUML.txt
3   question: Which class is involved in a composition
      relationship with QAction and is part of the QEvent
      hierarchy?
4 assert:
5   - type: llm-rubric
6     provider: openai:gpt-5-mini
7     value: |
8       Evaluate how well the answer identifies the correct
      class and relationship:

```


lished modeling principles. These questions were constructed to align with the conceptual focus of the evaluation, targeting core structural understanding and relational interpretation. Table 3 gives an overview of the used models. Table 4 summarizes the counted elements in each model variant.

Table 3. UML Models Used in supplementary Test Set

Set	UML Model	# Questions	Source
A	q-D1	1	[37]
A	q-E1	2	[37]
A	q-G1	3	[37]
A	q-I2	3	[37]
B	Process Modeling Metamodel	2	[21]
B	Software Process Engineering Meta-Model	2	[21]
B	Geppetto Core Meta-Model	2	[21]
Total		15	–

Table 4. UML Models and their descriptive statistics. C = # of classes, As = # of associations, S = # of structs, I = # inheritance, Ag = # aggregations, Co = # of compositions, Ab = # of abstract classes, E = # of enumerations, Att = # of attributes, M = # of methods

Model	C	As	S	I	Ag	Co	Ab	E	Att	M
qt-D	20	19	0	6	1	2	4	0	2	50
qt-D x2	62	33	5	6	1	16	4	5	48	151
qt-D x10	320	119	30	6	1	102	4	30	992	1250
qt-E	24	23	0	13	0	2	0	0	18	50
qt-E x2	61	38	4	13	1	16	0	5	70	125
qt-E x10	324	123	30	13	0	102	0	30	1008	1250
qt-G	28	20	0	8	1	0	2	0	1	42
qt-G x2	73	38	4	8	2	17	2	5	64	129
qt-G x10	328	120	30	8	1	100	2	30	991	1242
qt-I	33	17	0	9	0	0	3	0	7	67
qt-I x2	78	32	3	9	1	14	3	3	52	166
qt-I x10	333	117	30	9	0	100	3	30	997	1267
Process Modeling MetaModel	87	113	0	84	0	29	0	0	218	0
Software Process Engineering Meta-Model	87	113	0	84	0	29	0	0	218	0
Geppetto Core Meta-Model	95	176	0	78	0	75	9	0	80	0
Total	1953	1101	136	354	9	604	36	138	4766	5789

Figure 5 illustrates the results produced by promptfoo, containing test cases, model inputs, and responses from the evaluated LLMs, for one run. The figure also highlights the pass/fail classifications per test case, providing an overview of performance across the different providers and model configurations. If a question is classified as failed by the evaluation LLM, an explanation is also provided why it has failed and what was missing.

Furthermore, promptfoo offers a detailed insight into computational resource usage at test execution level. Figure 6 shows the token usage summary, including

question	uml	[google:gemini-2.5-flash-lite] You are an UML diagram assistant. Always ground your answers only in the provided UML Diagram. Format lists of classes or methods clearly. Keep your answers short. Question= {{question}}, UML Diagram= {{uml}}	[gemini:gemini-2.5-pro] You are an UML diagram assistant. Always ground your answers only in the provided UML Diagram. Format lists of classes or methods clearly. Keep your answers short. Question= {{question}}, UML Diagram= {{uml}}	[gemini:gemini-4o-mini] You are an UML diagram assistant. Always ground your answers only in the provided UML Diagram. Format lists of classes or methods clearly. Keep your answers short. Question= {{question}}, UML Diagram= {{uml}}	[gemini:gemini-5-mini] You are an UML diagram assistant. Always ground your answers only in the provided UML Diagram. Format lists of classes or methods clearly. Keep your answers short. Question= {{question}}, UML Diagram= {{uml}}
You need to improve the design of the module shown in the following diagram and remove sub-classes. What method will most probably need to be defined in QImageDevice from it's subclasses? (Note that the method can still be reimplemented in QImageDevice...)	<pre>@startuml abstract class QImageDevice { # paintEngine() : QPaintEngine! + width() : int + height() : int + numColors() : int } abstract class QImageIOHandler { + setDevice(device : QImageDevice!) + setFormat(format) + read(...)</pre>	[PASS] The method that will most probably need to be defined in QImageDevice from its subclasses is 'readData()'. This field should be protected.	[PASS] Based on the provided UML diagram, the method that should be moved to QImageDevice is 'readData()'. This method is currently defined in the following subclasses of QImageDevice: * 'QBuffer' * 'QFile' * 'QProcess' * 'QAbstractSocket'	[PASS] In the provided UML Diagram, the method that will most probably need to be moved to QImageDevice from its subclasses to remove redundancy is: - 'readData()' This method can be defined in 'QImageDevice' with a default implementation, allowing it to...	[PASS] Most likely you should move the low-level I/O primitives into QImageDevice so they can be shared by the other methods to add to QImageDevice (define here: subclasses reimplement): - protected readData(data : char, maxlen) : long - protected writeData(data : ...
You need to extend Qt's image plugin to include an image plugin to read and write images you invented, which class do you need to subclass for this plugin?	<pre>@startuml abstract class QImageDevice { # paintEngine() : QPaintEngine! + width() : int + height() : int + numColors() : int } abstract class QImageIOHandler { + setDevice(device : QImageDevice!) + setFormat(format) + read(...)</pre>	[PASS] You need to subclass 'QImageIOPlugin'.	[PASS] Based on the UML diagram, you need to subclass the following abstract classes: * 'QImageIOPlugin' This class is the plugin interface, with methods like 'create()' and 'capabilities()'. * 'QImageIOHandler' This class provides the imp...	[PASS] To extend Qt's functionality, you need to subclass the following class: - 'QImageIOPlugin' This class is the appropriate base for creating image IO plugins in Qt.	[PASS] You should subclass QImageIOPlugin. Recommended minimal classes/methods to implement: - QImageIOPlugin (subclass) - capabilities() - create() - keys() - QImageIOHandler (subclass returned by create()) - setDevice(device : QImageDev...

Fig. 5. Promptfoo Execution Results

Token Usage Summary:

Evaluation:
 Total: 42,283
 Prompt: 0
 Completion: 0
 Cached: 42,283
 Reasoning: 8,155

Provider Breakdown:
 google:gemin-2.5-pro: 16,450
 (16,450 cached, 8,155 reasoning)
 openai:gpt-5-mini: 10,210
 (10,210 cached)
 google:gemin-2.5-flash-lite: 7,988
 (7,988 cached)
 openai:gpt-4o-mini: 7,635
 (7,635 cached)

Grading:
 Total: 13,214
 Cached: 13,214

Grand Total: 55,497 tokens

Fig. 6. Promptfoo Token Overview

the distribution across LLM providers for evaluation and grading. This information provides transparency regarding execution efficiency and supports a better understanding of the operational overhead linked with model-based evaluation.

3 Results analysis

The main study relies on descriptive statistics, rather than inferential statistics. However, with the limited number of repetitions ($n=5$) in the experimental setup the data cannot be assumed to be normally distributed, violating the normality assumption of ANOVA. Non-parametric alternatives for ANOVA (that do not require normal distribution) are the Friedman test (for comparing at least 3 groups) or the Wilcoxon signed-rank test (for comparing 2 groups), however, these non-parametric test might not detect all significant effects that an ANOVA test (with a larger, normally distributed sample) can detect. Non-significant results should therefore be interpreted with caution, as they do not provide evidence for the absence of an effect.

The same analysis was performed for the results of the gpt-5-mini evaluator and the gemini-2.5-flash-lite evaluator. We obtained the same results for both evaluators. For RQ2.1, the difference between formats is analysed with a Wilcoxon signed-rank test for each size combination and threshold (e.g. the results for PlantUMLx2 are compared to the results for JSONx2 at threshold 25%). There are no statistically detectable differences for any of the combinations. For RQ2.2, the difference between model sizes is analysed with a Friedman test for each format and threshold, using the Nemenyi post-hoc test for pairwise comparisons. There is a statistically significant difference between size x10 and the baseline model size for both PlantUML and JSON. For RQ2.3, the difference between set A and set B is analysed with a Wilcoxon signed-rank test for each size of set A combined with set B, and each threshold (e.g. the results for PlantUMLx2 are compared to the results for PlantUML of set B at threshold 25%). There are no statistically detectable differences for any of the combinations. For

RQ2.4, the difference between LLM providers is analysed with a Friedman test for each set, using the Nemenyi post-hoc test for pairwise comparisons. In set A, there is a statistically significant difference between gpt5-mini and gemini-2.5-flash-lite, and between gpt5-mini and gpt4o-mini.

These analyses—based on the aggregated results per run—provide a first indication of statistically significant results. In future work, we plan to extend the experimental set-up to ensure that the data is tailored to more powerful parametric tests to further investigate the research questions.

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